III B.Tech II Semester

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23A39604	**ML MODEL OPTIMIZATION LAB (Professional Core)**	**0**	**0**	**3**	**1.5**

Course Objectives:

- To provide practical exposure to model optimization techniques for improving performance and efficiency of machine learning models.
- To explore various hyperparameter tuning methods and optimization algorithms.
- To apply regularization techniques to control overfitting and improve generalization.
- To evaluate model performance through various metrics and validation strategies.
- To learn pruning, quantization, and deployment strategies for optimized models in real-world applications.

Course Outcomes:

- Students will gain hands-on experience in optimizing machine learning models using advanced techniques like hyperparameter tuning, regularization, pruning, and quantization.
- Student will be able to assess model performance critically using evaluation metrics and implement strategies for reducing computational complexity.
- Students with the skills necessary to build efficient, scalable, and high-performing ML systems suitable for deployment in constrained environments such as edge devices and mobile platforms.

List of Lab Experiments:

1. Implementing Grid Search and Random Search
    - Hyperparameter tuning using Scikit-learn's GridSearchCV and RandomizedSearchCV
      (Cognitive Level: Apply & Analyze)
2. Bayesian Optimization using Hyperopt / Optuna
    - Optimize model hyperparameters using probabilistic approaches
      (Cognitive Level: Analyze & Evaluate)
3. Early Stopping in Training Deep Learning Models
    - Implement early stopping to avoid overfitting with TensorFlow/Keras
      (Cognitive Level: Apply & Evaluate)
4. Regularization Techniques (L1, L2, Dropout)
    - Apply different regularization methods to improve model generalization
      (Cognitive Level: Analyze)
5. Cross-Validation Techniques
    - K-Fold, Stratified K-Fold, and Leave-One-Out Cross-Validation
      (Cognitive Level: Apply & Evaluate)
6. Model Performance Evaluation
    - Use confusion matrix, precision, recall, F1-score, AUC-ROC for performance
      (Cognitive Level: Evaluate)
7. Model Pruning
    - Prune unimportant weights in a trained neural network using PyTorch/TensorFlow
      (Cognitive Level: Create & Evaluate)
8. Quantization Aware Training (QAT)

- Train a quantized model to reduce memory usage without losing accuracy **(Cognitive Level: Apply & Create)**

9. Knowledge Distillation
   - Transfer knowledge from a large model to a small one (student-teacher model) **(Cognitive Level: Analyze & Create)**
10. Optimizing Model Inference Time

- Use ONNX, TensorRT, or OpenVINO for faster model inference **(Cognitive Level: Apply & Evaluate)**

11. Automated Machine Learning (AutoML)

- Use Auto-Sklearn, Google AutoML, or TPOT for full pipeline optimization **(Cognitive Level: Evaluate & Create)**

12. Model Deployment Optimization

- Optimize model for deployment using TFLite/ONNX in edge devices **(Cognitive Level: Create & Evaluate)**

Text Books:

1. Aurélien Géron, **Hands-On Machine Learning with Scikit-Learn, Keras & TensorFlow**, O'Reilly.
2. Sebastian Raschka, **Python Machine Learning**, Packt Publishing.
3. Francois Chollet, **Deep Learning with Python**, Manning.

Reference Books:

1. Ian Goodfellow, Yoshua Bengio, Aaron Courville, **Deep Learning**, MIT Press.
2. Trevor Hastie, Robert Tibshirani, Jerome Friedman, **The Elements of Statistical Learning**.
3. Vijay Madisetti, **Machine Learning and Optimization Models for Real-Time Applications**, Springer.

Online Courses:

1. *Model Optimization Techniques – Coursera (DeepLearning.AI)*
2. Hyperparameter Tuning in Python – Udacity